

Not So Simple Voltage and Current Division Procedure

Instructional Objectives

1. Familiarize students with series and parallel equivalent resistances.
2. Predict and measure circuit quantities in circuits with series and parallel components.
3. Design a circuit to deliver a specified voltage according to given constraints.
4. Learn how to use the Oscilloscope.

If you don't recognize the material in this lab then look at the Theory manual for this lab to remind you or bring you up to speed on this subject.

Procedures

0) Parts list: All 1/4W resistors. 1-1.5K, 1-3.3K, 1-6.2K, 1-6.8K, 2-8.2K, 1-20K, 1-30K, 1-33K.

1) Complicated resistor network analysis and measurement:

- a) Measure the values of the resistors using the Digital Multi-Meter (DMM). Record the values of R_1 to R_6 . Remember your fingers will affect the measured resistance!

$R_1 =$ 1.508 $\text{K}\Omega$, $R_2 =$ 29.814 $\text{K}\Omega$, $R_3 =$ 6.7927 $\text{K}\Omega$, $R_4 =$ 3.2996 $\text{K}\Omega$, $R_5 =$ 6.1307 $\text{K}\Omega$, $R_6 =$ 19.992 $\text{K}\Omega$

b) Build the circuit shown in Figure 1.

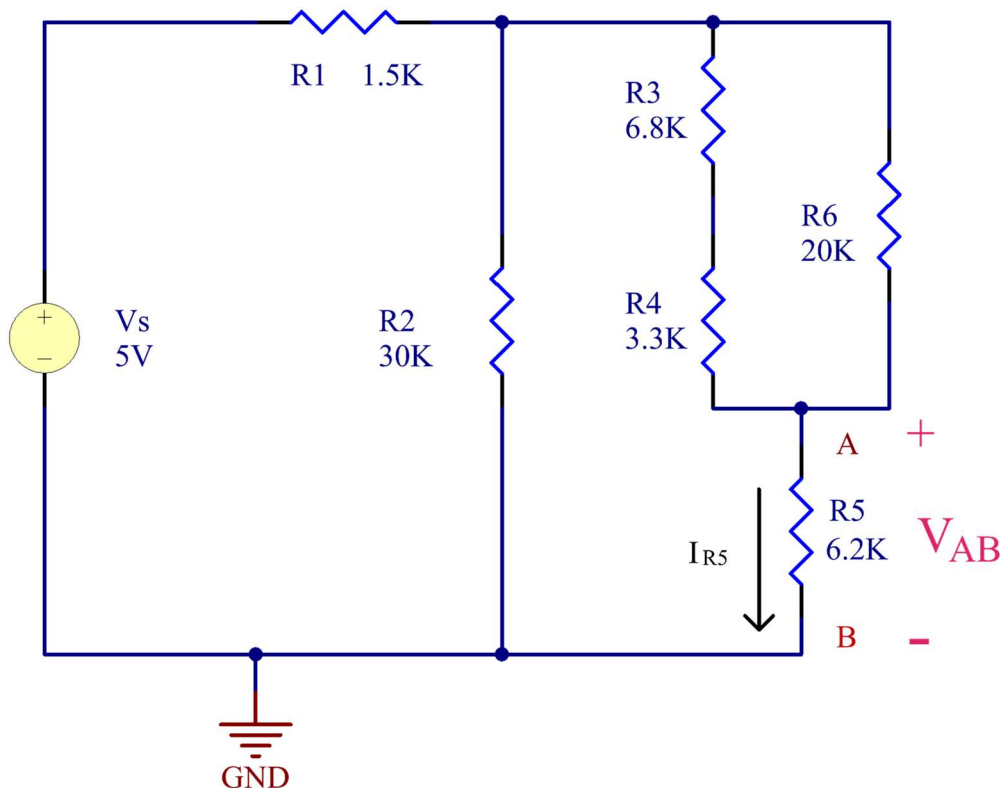


Figure 1: Not So Simple resistor circuit.

Set the YELLOW DC Power Supply to supply 5V DC with a 25mA current limit.

- Connect the red terminal of the YELLOW Power Supply to R_1 on the breadboard.
- Connect the black terminal of the YELLOW Power Supply to your GND node on the breadboard.

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- c) Measure the voltage output of the power supply using the DMM $V_S = \underline{5} \text{ V}$.
- d) Use the DMM to measure the voltage drop from point A to point B as well as the current through resistor R_5 (I_{R5}). The procedure to measure current using DMM was covered in Lab 1.

$$V_{AB} = \underline{2.044 \text{ V}} \quad I_{R5} = \underline{0.584 \text{ mA}}$$

- e) Use the values of V_{AB} and I_{R5} to calculate the equivalent resistance between points A and B.

$$R_{eq} = \underline{3.5} \text{ K}\Omega.$$

NOTE: If you put this circuit in the LTSpice simulator you can see what the voltages and currents should be. It is a good idea to learn how to use the simulator because it works well and you will be able to use it on the bench exam at the end of the semester if you want to.

- 2) KVL and KCL on a series parallel circuit.

- a) Get 2 - 8.2K for R_1 and R_3 and a resistor between 20K and 39K for R_2 .

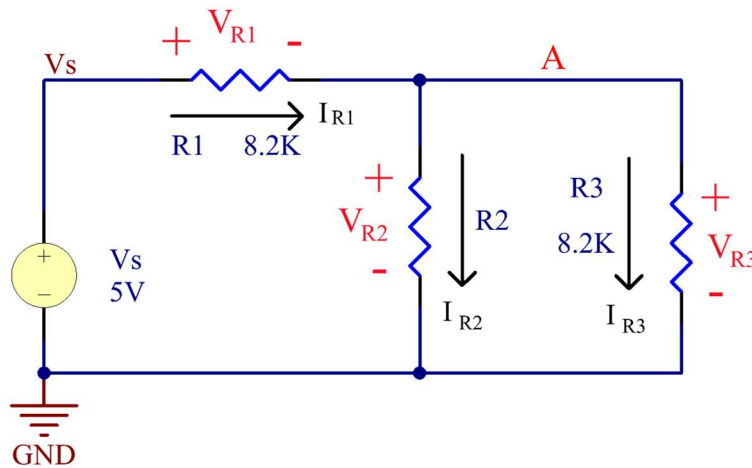


Figure 2

- b) Build the circuit in Fig. 2. Connect the YELLOW Power Supply to R_1 and GND.
- c) Measure the voltage drop across each resistance (V_{R1} , V_{R2} and V_{R3}) with the DMM. Enter the data in Table 1.
- d) Use the measured value of resistance and voltage to "Calculate" I_{R1} , I_{R2} and I_{R3} . ($I=V/R$). Enter the calculated values of current in Table 1.
- e) Determine the values of the current using $I=V/R$ and enter them in the "Theoretical Current" column. (Use the nominal resistor values and measured voltages)

Resistor	Nominal R Value (k Ω)	Measured R Value (k Ω)	Measured Voltage (V)	Calculated Current (mA)	Theoretical Current (mA)
R_1	8.2	8.193	2.79	0.341	0.3415
R_2	30	29.847	2.208	0.074	0.0733
R_3	8.2	8.118	2.208	0.272	0.2682

Table 1: Measured and Calculated Data for Fig. 2.

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f) Measure the voltage output of V_s (the power supply) using the DMM: $V_s = \underline{4.9997 \text{ V}}$.

Calculate the % error of your "Calculated Current" values compared to the "Theoretical Current" values. Use the "Calculated Current" values from above in your second column and use the Theoretical values as the reference values for the third column in the % error calculation. Place these results in Table 2.

Resistor	Calculated Current (mA)	Theoretical Current (mA)	Calculated vs Theoretical Error %
R_1	0.341	0.3415	0.1464
R_2	0.074	0.0733	0.955
R_2	0.272	0.2682	1.417

Table 2: Calculated Data and Errors for Fig. 2.

3) YOU CAN EARN EXTRA POINTS for your COURSE FINAL GRADE in this Task:

Here is a design problem. Your task is to design a circuit that will deliver 2.0V, or 2.5V, or 3.0V $\pm 5\%$ (Pick 1) across a load resistor of 8.2k Ω . Figure 3 and the following text presents the problem.

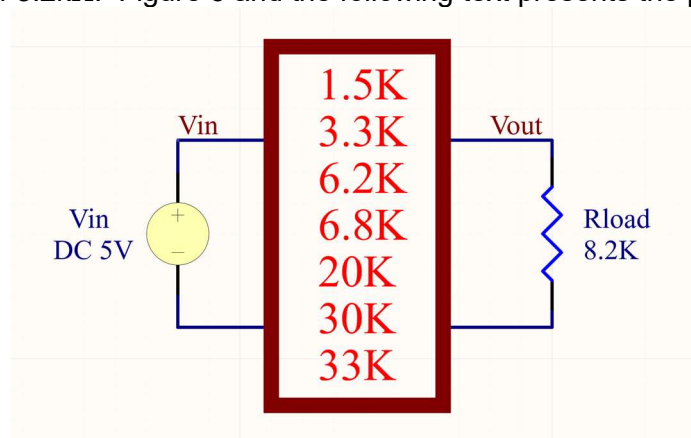


Figure 3: Circuit Design Schematic for Step 3

The resistors available for use are listed inside the box in Fig. 3. You can use more than one of any value. The resistors can be in series and/or parallel. There are design constraints as with most engineering design problems. In this case, your component cost must be less than 7 cents. You only need to pay for the resistors, which cost 2 cents each. The load resistor is free. Assume your time is free but only this one time ;-).

- Pick an output voltage across R_{LOAD} of 2.0, 2.5 or 3.0V. 3.0 V V_{OUT} .
- Design and document a circuit that will meet the design criteria. Document means a schematic.
- Record your proposed solutions and brainstorm until you have found a solution.

NOTE: The group that can drive the lowest value of R_{LOAD} to within 5% of your chosen voltage in simulation after your design is finished gets extra credit on the final grade. You still have to drive the 8.2K R_{LOAD} to the desired voltage.

- Construct your proposed circuit. Verify that it meets the design criteria by measuring V_{IN} (5V source) and V_{OUT} (V_{RLOAD}).
- When your circuit is working, demonstrate to the lab instructor/TA measured V_{in} and V_{out} .

Measured: V_{IN} 4.999 V V_{OUT} 3.0435 V

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Generate the Thevenin Equivalent circuit for your design. R_{TH} 1030 Ω V_{TH} 3.44 V

Draw the equivalent circuit here (No R_{LOAD})



Determine the lowest value of R_{LOAD} your designed circuit can drive and still get the desired output voltage within 5%. Make sure it still gets the correct output voltage for the 8.2K Ω load. **The group or groups that get the lowest resistor value that can be driven gets points added to your course final grade. There will be 3 awards. One for each desired voltage.**

I found the lowest value of R_{load} was 4980 Ohms, at this value, the equivalent circuits drove 2.85 Volts through the load resistor. This is right at the edge of the 5% threshold.

Learn the Oscilloscope.

The oscilloscope allows you to look at signals in time. So as a sine wave varies over time you can see the sine wave on the screen of the oscilloscope.

Usually an Oscilloscope is used to measure AC waveforms. A DVM can be used but you don't get to see the waveform shape. The DVM gives a voltage value and it may be correct but you don't know if the waveform is distorted or misshaped somehow. The Oscilloscope also has a Signal Generator built into it. This can supply some DC and many AC waveforms. This generator is usually used to measure many circuits as you will find out as we go forward.

Before we change over to the AC source and look at it with a scope I want to point out that there are many ways to quantify AC voltage. Some of the more common are volts peak, volts peak to peak and volts RMS. These are all used and you should understand the differences.

Volts peak, V_P , AC only amplitude. V_P is used for quantifying the amplitude of a symmetrical waveforms like a sine wave where the top half of the waveform is the same size and shape of the bottom half. In this case specifying the peak voltage tells you all you need to know about the amplitude. Many signal generators use V_P for the amplitude of symmetrical waveforms.

Volts peak to peak, V_{PP} , AC only amplitude. V_{PP} is used to quantify any waveform symmetrical or asymmetrical. It is a value which tells the voltage between the maximum and minimum of the waveform. It does not indicate the DC value (or offset) of the waveform.

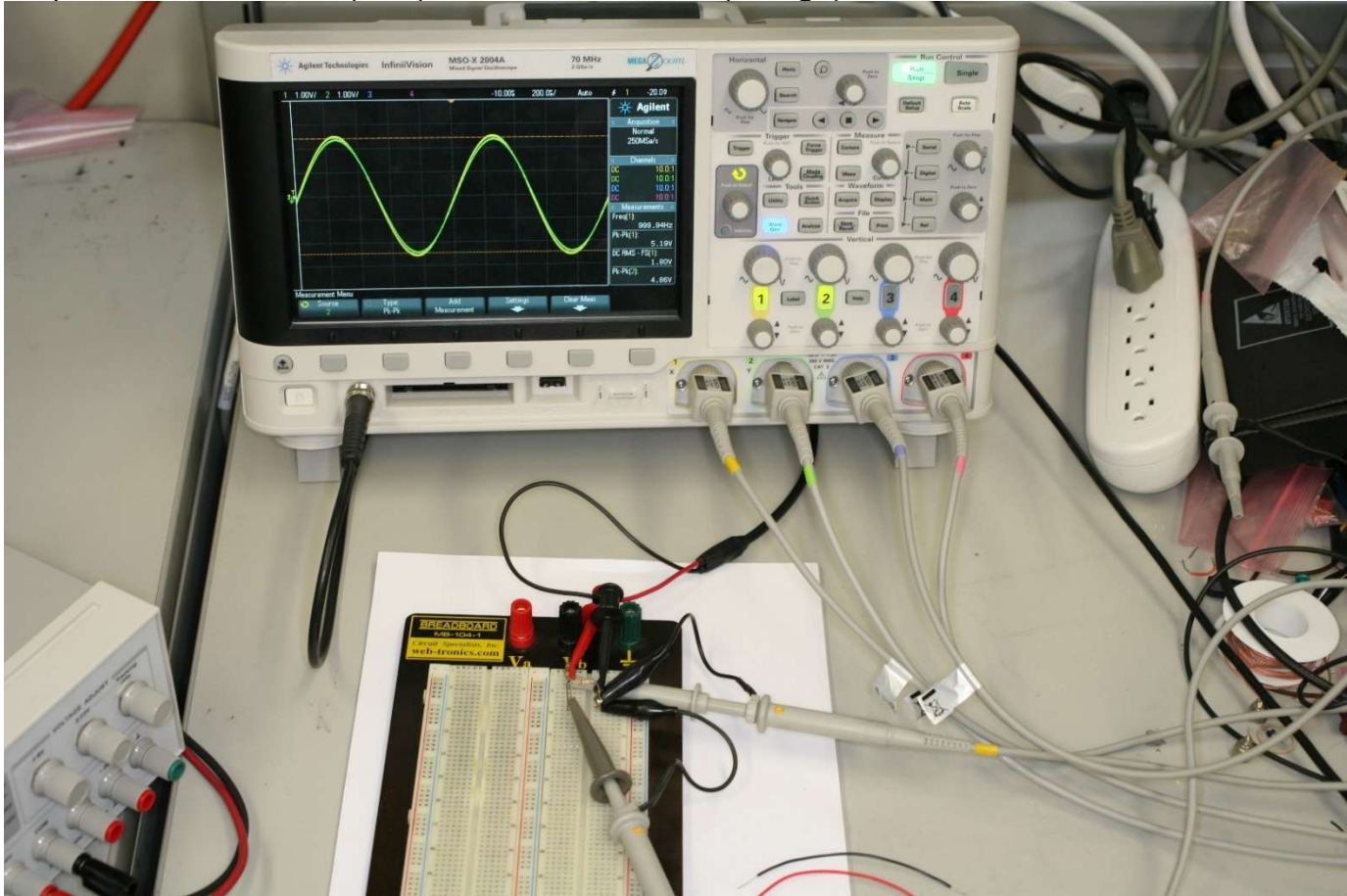
AC RMS, tells the RMS value of only the AC portion of the waveform. It ignores the DC component of the waveform. You learned in ECE230 that the RMS value gives the voltage equivalent in DC of an AC signal. The equivalence is the DC signal and the AC RMS signal will dissipate the same power in a 1 Ω resistor.

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RMS, tells the RMS value of a waveform considering both AC and DC components of the waveform. To get RMS you have to take the $\text{SQRT}[(\text{AC RMS})^2 + \text{DC}^2]$

Finally **DC**, gives the offset of any AC signal. This is often 0. For example when you have a sine wave that goes from 1V to -1V the energy above 0 is equal to the energy below 0 yielding an offset of 0. A 2V_{PP} with 1V offset would yield a sine wave that went from 0V to 2V. This has an average DC value of 1V. Thus the 1V offset.

The scope we use has 4 channels. They all work the same way. The 2 major things to setup on the scope are the horizontal (time) axis and the vertical (voltage) axis.



The largest knob at the top is used to adjust the time that shows on the screen. For example you can set the time across the screen to 1sec, 10mS or even 50nS. The 4 large knobs near the center of the scope are for adjusting the Volts/division of each channel of the 4 channels. Divisions are drawn onto the screen as a grid. Both horizontal and vertical divisions. The range for volts per division is 1mV to 5V per div. Notice in the picture there are scope probes plugged into the input BNC connectors. Using probes is almost always the way signals are measured.

You are going to measure a couple signals today so you can see how to make an accurate useful measurement.

Before we measure anything, I want to tell you that anytime you connect measurement equipment to a circuit you are actually connecting an impedance to the point you are measuring in the circuit. What you want to connect is an infinite impedance to the circuit so the circuit is not changed when you measure it. Infinite is not possible. We get what the instrument gives us. Our oscilloscopes have a 1M Ω input resistance with a small

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amount of capacitance. This is what we get when we use a probe that is a X1 scope probe. This info is written on the scope front panel. $1\text{M}\Omega$ is OK if you are connecting the scope to a $10\text{K}\Omega$ or less resistance. If you need more impedance there is the option of using a X10 scope probe. It gives you $10\text{M}\Omega$ input resistance with a 10 times reduction in the signal amplitude. Two problems with 10X probes is they need to be "Calibrated" and the signal is 10 times smaller than the X1 scope probe.

Most of the measurements you will make in ECE270 can be made with a X1 scope probe.²¹

I want you to setup the scope and input signals exactly as stated. This is to help you learn 'good' and 'not so good' ways to measure signals. In general, you want to make the vertical signals as large as possible without clipping them off at the top or bottom.

- Connect a scope probe to the CH1 (Yellow) input. CH1 is on if the number 1 below the Sensitivity knob is illuminated.
- Make sure the probe switch is set to X1.
- Set the V/Div Sensitivity to 1V/div on CH1.
- Set the time/Div to 200uS.

This doesn't matter much for DC. It is much more important for AC signals. You have options for nS/Div to seconds/Div for time.

There is a small ground symbol on the left side of the screen colored yellow.

- Put it in the center of the screen.

You can do this by either turning the small position knob under the illuminated 1 or by pushing the position knob in.

- Set the power supply (Yellow) channel to 1.000V.
- Measure this DC voltage with the DMM. 1.00 VDC Measure AC as well. 0.001 VAC.

These measurements should be 1.000VDC and 0VAC. The measurements should close to this but usually not exactly this.

- Connect a Banana to mini clip to the Red and Black output jacks.
- Connect the Scope probe to the power supply output. Scope probe tip to the wire coming from the red jack and the ground clip to the wire coming from the black jack.
- You should see a straight yellow line going across the screen 1 division above the ground symbol.

The scope can measure the voltage of this line.

- Push the Meas button for you.
- There are about 50 things you can measure which should show up when you push the Meas button.
- Pick Average Full Screen to measure the value of the DC signal of channel 1. It should be something like 1V. Avg-FS(1) 1.00 V.
- Add Peak to Peak for channel 1. It should be 0V for DC. Pk-Pk(1) 1.2 mV.

The trace on the screen is probably a little noisy. You can do things to reduce this noise. Reducing the noise is good as it improves the accuracy of the measurement. More on this soon.

- Change the Sensitivity to 5V/Div. Measure again Avg-FS(1) 1 V. Pk-Pk(1) 0 mV.
- Change the Sensitivity to 2V/Div. Measure again Avg-FS(1) 1 V. Pk-Pk(1) 0 mV.
- Change the Sensitivity to 500mV/Div. Measure again Avg-FS(1) 1 V. Pk-Pk(1) 0 mV.
- Change the Sensitivity to 200mV/Div. Move the GND reference down until the trace is on the screen. Measure again Avg-FS(1) 1 V. Pk-Pk(1) 0 mV.

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To improve accuracy using the scope Meas function you can do several different things. The first thing you can do is press the Acquire key and pick the High Rezolution option of acquisition. Try this now with 0.2V/div. This changes the number of bits used to convert the analog signal to digital from 8 to 12 bits. Try this now with 0.2V/div. Avg-FS(1) 1 V. Pk-Pk(1) 0 mV.

Another thing you can do is choose Averaging in the Acquire menu. This averages X number of screens of data. X is chosen with another soft key. Every doubling of averages improves the noise by a factor of 2. Pick 32 as the number of averages. Remeasure:

5V/div Avg-FS(1) 1 V. Pk-Pk(1) 0 mV.
2V/div Avg-FS(1) 1 V. Pk-Pk(1) 0 mV.
1V/div Avg-FS(1) 1 V. Pk-Pk(1) 0 mV.
500mV/div Avg-FS(1) 1 V. Pk-Pk(1) 0 mV.
200mV/div Avg-FS(1) 1 V. Pk-Pk(1) 0 mV.

You should have seen a great improvement in the Pk-Pk(1) measurement at all V/div settings.

Push Auto scale with the V/div set to 0.2V/div.

Several things happened. Both are bad. The V/div probably changed to a small number and the time /div is now real fast. The reason you don't want these setting is because as soon as you go to something that is not DC you won't see useful traces on the scope.

- Connect the probe to the output of the Wavegen using a BNC to mini clip cable on the wave gen output.
- Set the Wave Gen 1 VPP Sine with 0V offset at 1KHz.
- Push the position button for CH1 to set the GND marker to center screen. You want center screen because the Sine wave has 0V offset.

If Autoscale set the time / Div in the nS range and the horizontal to some mV/Div range you probably won't see anything.

- With a 1KHz sine wave set the time to 200uS/Div and the Horizontal to 200mV/Div. Most likely you see a sine wave that is moving around the screen. Sometimes not. This is determined by the Triggering function of the scope.
- For now set Triggering to Edge type on CH1 with a positive slope.
- Then adjust the Level knob so that the level is somewhere between 0.4V and -0.4V.

This should make the image on the screen hold still. Another thing you can do is:

- Press the level button and the scope will pick a good spot to set the level to.
- The BEST thing to do for triggering when you are using the Sig Gen to supply the signal to the circuit is to set Triggering Source to Wavegen. This will always trigger the scope successfully.

Do this now so you can do the next step.

Almost Done. You are going to do the same thing for the sine wave that you did with the DC.

- Make the sig gen is putting out a 1V_{PP} 1KHz 0V offset sine wave.

Make the measurements like before. Start with the Acq mode type to Normal. Use Avg N Cycles instead of Avg Fullscreen.

5V/div Avg-Cyc(1) 0 V. Pk-Pk(1) 1000 mV.
2V/div Avg- Cyc(1) 0 V. Pk-Pk(1) 1000 mV.
1V/div Avg- Cyc(1) 0 V. Pk-Pk(1) 1000 mV.
500mV/div Avg-Cyc (1) 0 V. Pk-Pk(1) 1000 mV.
200mV/div Avg-Cyc (1) 0 V. Pk-Pk(1) 1000 mV.
100mV/div Avg-Cyc (1) 0 V. Pk-Pk(1) 1000 mV.
50mV/div Avg-Cyc (1) clipped V. Pk-Pk(1) clipped mV.

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You should notice that for 50mV per div you didn't really get good numbers where as for 100mV/Div and more resulted in good numbers.

What is going on here is that when the scope traces clip a lot the measurement can't work because the data is off screen. A small amount of clipping is OK. The measurements will let you know with the value is clipped or the value is >XXX V. When that happens you need to make the waveform smaller on the screen.

Repeat the measurements with the Acquisition mode set to Averaging.

5V/div Avg- Cyc(1)	<u>0</u>	V. Pk-Pk(1)	<u>1000</u>	mV.
2V/div Avg- Cyc(1)	<u>0</u>	V. Pk-Pk(1)	<u>1000</u>	mV.
1V/div Avg- Cyc(1)	<u>0</u>	V. Pk-Pk(1)	<u>1000</u>	mV.
500mV/div Avg- Cyc(1)	<u>0</u>	V. Pk-Pk(1)	<u>1000</u>	mV.
200mV/div Avg- Cyc(1)	<u>0</u>	V. Pk-Pk(1)	<u>1000</u>	mV.
100mV/div Avg- Cyc(1)	<u>0</u>	V. Pk-Pk(1)	<u>1000</u>	mV.
50mV/div Avg- Cyc(1)	<u>clipped</u>	V. Pk-Pk(1)	<u>clipped</u>	mV.

You should see a slight improvement with averaging compared to Normal Acquisition. So where can you see a larger improvement?

- Change to Wave Gen amplitude to 20mVPP.
- Go back to Normal mode of Acquisition.
- Change the V/Div to 5mV

You should see a sine wave that is fuzzy, has a lot of noise.

Use the Averaging Acquisition mode. You should see a significantly quieter (Less noisy) waveform. The VPP should be closer to 20mV too.

Best practices: Use Triggering on the Wave Gen. Make the AC signals as close to full screen in size as you can. Make DC signals near the Top or Bottom of the screen with the GND at the opposite edge of the screen. Use a Time/Div that is near the period of the AC waveform you are measuring.

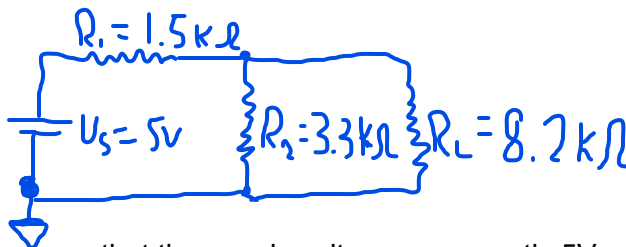
Post Lab Questions

1. Compare the "Theoretical Currents" to the "Measured Currents" for Fig. 2. Explain why they may not be exactly equal. Verify that Kirchhoff's Current Law applies to this circuit.

The values may be off a bit from the theoretical value because of analog noise, improper resistor values, and measurement uncertainty. I can verify that KCL holds for this circuit I.E $I_{R1} = I_{R2} + I_{R3}$: $0.341 \text{ A} = 0.074 \text{ A} + 0.272 \text{ A}$.

2. For Figure 2, verify KVL for the loop containing V_s , R_1 and R_2 . Use the voltage values that you measured in step 2 of this lab. The KVL equality, that is $V_s = V_{R1} + V_{R2}$, does hold: $5\text{V} = 2.79\text{V} + 2.208\text{V}$.

3. Draw (by hand or by program) the circuit you designed in step 3. Explain the reasoning you used to get to your final solution and discuss how you verified that the circuit met the design criteria. I recommend you use LTSpice to draw the circuit.



Essentially all I did was place the $R_1 = 1.5\text{k}\Omega$ resistor in series with the load resistor in parallel with R_2 and tested various values of R_2 until I measured a desirable voltage.

4. Assume that the supply voltage was exactly 5V and the resistors were ideal in step 3. What voltage did you choose to put out. For what range of R_{LOAD} will your circuit deliver the voltage you chose $\pm 5\%$? In other words, what is the maximum and minimum resistance of R_{LOAD} for which the circuit will still operate as specified?

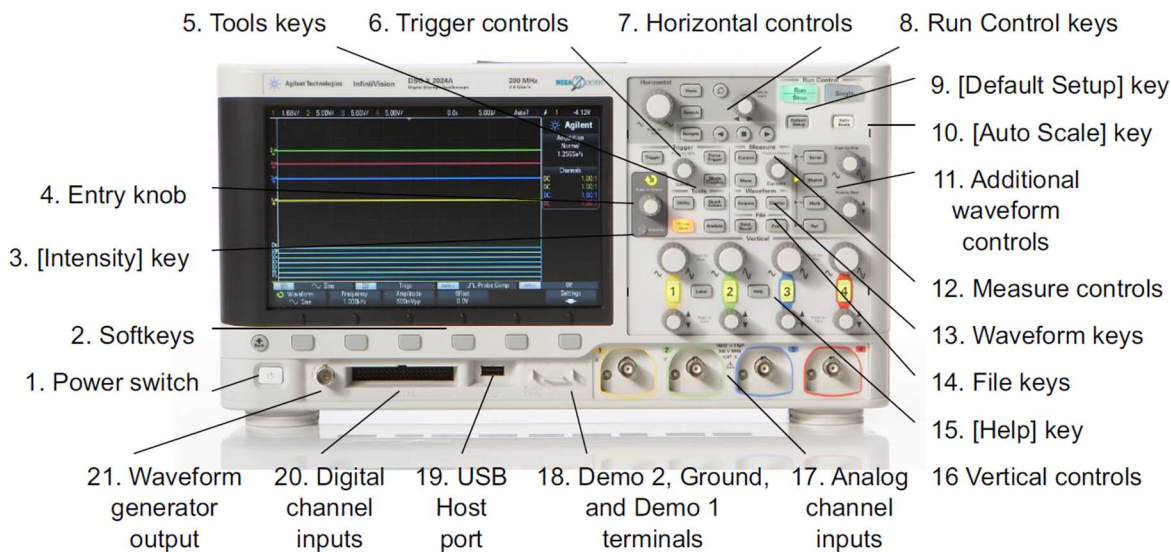
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Earlier, I chose V_{out} to be 3V calculated the minimum value of R_{load} to be 4980 Ohms. At this value V_{out} was at 2.85V, right at the lower threshold. In a similar way, $V_{th} \cdot (R_L / (R_{th} + R_L))$, I was able to calculate the maximum load resistor value to be about 11188 Ohms. At this value, V_{out} is 3.15V, right at the top of the threshold.

5. Refer to Figure 1. Give the currents through R_1 , R_2 , R_3 , R_5 and R_6 when $V_s = 10V$ and $R_1 = R_2 = R_3 = R_4 = R_5 = R_6 = 1K\Omega$.

I_{R1}	I_{R2}	I_{R3}	I_{R5}	I_{R6}
6.15 mA	3.85 mA	0.769 mA	2.31 mA	1.54 mA

Appendix: Simple description on using the Keysight MSO-X 2004A Oscilloscope.
First the front screen.



The Softkeys and Entry Knob will get used a lot when running the scope. Let's define the leftmost softkey as key #1 with the rightmost as key #6.

Setting up the measurement channels: Vertical controls are used to turn on and setup the measurement channels. Any channel that is enabled will have the number lit up as the picture shows. Depress the number button of the channel you want to adjust. A menu will show up above the softkeys. #1 sets the coupling of the input to the probe. AC means all DC signals will be blocked by a series capacitor built into the scope. DC coupling allows AC and DC signals to pass into the scope. We usually use DC coupling. #4 sets fine or course range adjustment. Course adjustment sets the volts per divisions in a 1-2-5-1 sequence. 1V/div, 2V/div, 5V/div etc. This is usually good enough for most measurements. #6 is to set the probe value. We use 10:1 probes. The probe ratio is displayed on the right hand side of the screen. If they are not at 10:1 then use the probe softkey to adjust the setting to 10:1. This tells the scope what the voltage value is that is being measured. So for a 10:1 probe a 1V signal at the input is really a 10V signal at the probe tip.

Setting the Horizontal time base: For most of your measurements you will only use the large knob which changes the time per division. Once again in course mode the sequence is 1-2-5-1. The course or fine mode is toggled by pushing the large knob in the Horizontal group.

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Setting Triggering: Triggering determines when the scope draws the images on the screen. Press the “Trigger” bottom just above the Entry knob. This brings up 3 soft keys. #1 sets the Trigger Type. You will usually use ‘Edge’. #2 sets the input channel that will be used for the trigger signal. #3 sets the Slope of the trigger signal. You will usually use rising or falling slope. The small knob marked level is the 4th adjustment you will usually use. There are 2 modes of triggering. Auto and Normal. The mode is indicated in the upper right part of the scope display. In Auto mode the scope will watch for the trigger event you defined for example Rising Edge at 2 volts. If this event never happens the scope will trigger anyway. In Normal the scope watches for the event and only triggers when the event occurs. To set the trigger mode: press “Mode/Coupling” which is the button to the left of the ‘Entry’ knob. Softkey 1 selects Auto or Normal.

A description of what trigger does is useful here. The scope will display images of the waveform. When the display is drawn is determined by a trigger event. So for example in this lab we want to watch the voltage across the capacitor. This is measured with channel 2. The voltage across the capacitor is almost a square wave going from 0 to 4 volts and 4 to 0 volts. For convenience we will trigger the drawing of the images when the voltage crosses 2V. In one case we want the voltage to be rising from 0 to 4 when we trigger. This happens when the capacitor is being charged. In another case we want to trigger when the voltage is falling from 4 volts to 0. This happens when the capacitor is being discharged. So in this lab you set up the type as ‘Edge’ on channel 2 with a rising edge. Next all we need to do is change the trigger to a falling edge when we want to see the discharge. The push of one trigger button and the whole image changes.

Special Instructions for measuring DC only. Set the trigger type to Auto. That’s it.

Cursors: You can have up to 4 cursors on the screen. 2 horizontal (voltage) and 2 vertical (time). Press the Cursor button. 4 softkeys come up. #1 set to manual mode. #2 choose the channel that has the voltage you want to know the value of. #3 sets which cursor the ‘Entry knob’ will control. You will need to place X1, X2, Y1 and Y2 y choosing which one you want to move and placing them with the Entry knob. Once the 4 cursors are in place you can read the deltas between them on the screen.

Scope Probe Calibration: In the 10X setting the scope probe needs to be calibrated. What does this mean? Calibration using a very simple explanation allows the probe to measure the voltage correctly. The following description should help clarify how things can go wrong without a calibrated probe.



The scope has a built-in square wave output to make it easy to adjust the probe calibration. See picture to the left. Just connect the probe tip and black ground lead to the square wave test points and see what you get. You may have to adjust the V/Div and time/div to get a squarish looking signal. If the square wave is rounded or has a pointy overshoot (Fig 11a and b) then the probe needs to be calibrated.

To calibrate a probe use the square wave calibration output on the scope as shown to the left. It is built into the scope. Make sure the probe is set to X10. Calibration is only possible on the X10 probe setting. X1 does not need any calibration. Turn the trimmer in the hole on the scope probe base (near the BNC connector) with a plastic screwdriver that came with the probe. By turning the adjustment screw you should see the signal overshoot and undershoot (see figure 11). Adjust the calibration until the waveform looks like a square wave with no under or over shoot and measure the correct waveform voltage.

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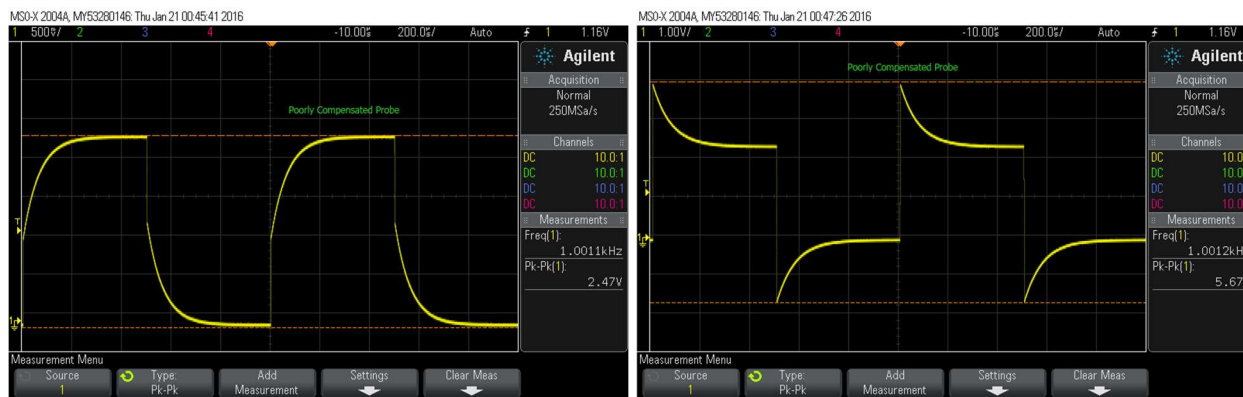


Figure 11a (undershoot or undercompensated) and 11b (overshoot or over compensated)

I STRONGLY SUGGEST you put the colored rings on the probes so it is easy to identify the channel number a certain probe is connected to. The colors match the oscilloscope channel colors used on the screen.